# Curriculum Vitae

**SAMEER MALIK**
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## Research Interests

**Observational cosmology; large-scale structure of the Universe.** My research uses galaxy clustering to constrain structure formation and cosmic expansion, with a focus on redshift-space power spectrum and bispectrum multipoles. I study observational systematics, particularly fiber collisions, in spectroscopic surveys such as the Dark Energy Spectroscopic Instrument (DESI), and use N-body simulations (AbacusSummit) together with Fisher forecasts to validate estimators and models for current and upcoming surveys.

## Education

**Indian Institute of Technology Kharagpur, Kharagpur, India** 2024 – Present
Ph.D. in Physics (CGPA: 8.69/10)
*Redshift Space Bispectrum as a probe of Large-scale structure*
Advisor: Prof. Somanth Bharadwaj

**Indian Institute of Technology Kharagpur, Kharagpur, India** 2022 – 2024
M.Sc. in Physics (CGPA: 8.74/10)
*Tidal force in Newtonian Gravity*
Advisor: Prof. Sayan Kar

**University of Delhi, Delhi, India** 2018 – 2021
B.Sc. in Physics (CGPA: 8.87/10)

***Ph.D. Supervisor**: Prof. Somnath Bharadwaj, IIT Kharagpur*
***Ph.D. Collaborator**: Dr. Shadab Alam, TIFR Mumbai*
***M.Sc. Supervisor**: Prof. Sayan Kar, IIT Kharagpur*

## Research Experience

**1. Constraining Early Dark Energy using CMB, BAO, DES Y5 and Pantheon+ — Prof. Elisabeth Krause (University of Arizona), Dr. Tanvi Karwal (University of Chicago) & Dr. Anoma Ganguly (University of Arizona)[link] 27 Jul – 4 Aug 2026**

- Analyzed **CMB power spectra** and **BAO** measurements from galaxy catalogs to assess their sensitivity to fundamental physics.
- Applied **Bayesian MCMC** inference to place constraints on early- and late-time dark energy models.
- Carried out complementary **frequentist profile-likelihood** analyses to cross-validate the dark energy parameter constraints.
- Examined the degeneracy-breaking power obtained by jointly combining CMB and BAO expansion-history measurements.

- Incorporated **Pantheon+ Type Ia supernova** data to further constrain dynamical dark energy scenarios.

**2. Analytical Models for Gravitational Collapse — Dr. Sayan Kar (IIT Kharagpur)**[link] **Jan – Apr 2025**

- Formulated analytic solutions describing spherically symmetric gravitational collapse within the framework of **General Relativity**.
- Characterized the physical conditions governing black hole formation and the evolution of event horizons.

**3. Gravitational Instability in General Relativity — Dr. Sayan Kar (IIT Kharagpur)**[link] **Jan-Apr 2025**

- Investigated the onset and subsequent growth of gravitational instabilities within a fully relativistic framework.
- Established connections between relativistic fluid equations and the process of large-scale structure formation.

**4. Comparison of K-means and Gaussian Mixture Models using MNIST — CCDS, IIT Kharagpur** **Aug – Dec 2024**

- Implemented and rigorously evaluated **K-means** and **Gaussian Mixture Model (GMM)** clustering algorithms on the MNIST dataset.
- Analyzed cluster purity and visualized latent-space representations to compare the separability achieved by each method.

**5. Image Filter Application using PIL — Dr. S. K. Reddy (Computational & Data Sciences, IIT Kharagpur)**[link] **Aug – Dec 2024**

- Developed a Python-based **graphical user interface** for applying a suite of image-processing filters using the `PIL` library.
- Implemented blur, sharpen, grayscale, brightness adjustment, and custom convolution kernel filters.

**6. Violation of Active and Passive Mass Equality Using Lunar Laser Ranging — Dr. Sayan Kar (IIT Kharagpur)**[link] **Aug – Dec 2024**

- Tested the observational validity of the **equivalence principle** using precise Earth–Moon distance measurements obtained via lunar laser ranging.
- Explored the theoretical implications of a potential inequality between active and passive gravitational mass.

**7. Redshift Space Distortion & Selection Function for Redshift Surveys — Dr. Somnath Bharadwaj (IIT Kharagpur)**[link] **Aug – Dec 2024**

- Derived the relationship connecting real-space and redshift-space overdensities for observers in both the **CMB** rest frame and the **Local Group (LG)** frame.
- Established the link between real- and redshift-space **selection functions**, and implemented **Turner's method** to compute the selection function for a **flux-limited redshift survey**.

**8. Density Estimation using Deep Generative Neural Networks — Dr. Adway Mitra (Artificial Intelligence, IIT Kharagpur)**[link] **Aug – Dec 2024**

- Explored probabilistic density modeling techniques using **Variational Autoencoders (VAEs)** and **Normalizing Flows**.
- Benchmarked these generative models against one another for likelihood-based density estimation performance.

**9. Formation of Large Structures under Tidal Forces — Dr. Sayan Kar (IIT Kharagpur)**[link] **Aug 2023 – Jul 2024**

- Investigated the influence of **tidal forces** on gravitational structure formation within a Newtonian dynamical framework.
- Modeled density profiles using both **Isothermal** and **Plummer** models for collapsing systems.
- Derived and applied the **Jeans instability** criterion in the presence of external tidal effects.
- Performed a stability analysis on **142 fragments** arising from a collapsing density envelope.

**10. Optical Waveguides and Resonators - Dr. Shivakiran B. N. Bhaktha (IIT Kharagpur) July-2023 - Dec-2023**

- Fabrication of optical waveguides and resonators via UV and soft lithography.
- Developed stretchable, hexagonal microlasers.

**11. Projectile Motion with Quadratic Drag — Dr. Sayan Kar (IIT Kharagpur) May-2023 - July-2023**

- Analyzed **trajectory asymmetry** in 2D under drag forces.
- Studied **asymptotic behavior** and energy dissipation mechanisms.

## Teaching and Mentoring Experience

- Teaching Assistant for 2st Year B.Tech. Physics Lab, IIT Kharagpur (Jan–Apr 2025).
- Teaching Assistant for 1st Year B.Tech. Physics Lab, IIT Kharagpur (Aug–Dec 2025).
- Teaching Assistant for Astronomy and Astrophysics course under Professor Somanth Bharadwaj, IIT Kharagpur (Jan-Apr 2026).
- Mentoring M.Sc student Soumyadeep Mukherjee on his MTP project **"Higher order perturbation theory in cosmology"** at IIT Kharagpur (July- 2025 - present).

## Honors, Awards and Grants

- Received a **travel grant of €320** from ICTP, Italy, and Chulalongkorn University, Bangkok, to attend the Bangkok School on Cosmology and the Workshop on the Dark Side of the Universe, Bangkok, Thailand, 7–18 December 2026.
- Received a **grant of €195** for local expense from ICTP, Trieste, Italy, to attend the Summer School on Cosmology, 29 June–10 July 2026.
- Received a **travel grant of INR 60,000** from the International Relations Office, IIT Kharagpur, to attend the ICTP Summer School on Cosmology (2026).
- Awarded a **merit-based scholarship** (INR 1,000/month) during the M.Sc. at IIT Kharagpur for two years (2022–2024).
- Secured **All India Rank (AIR) 261** among 62,654 candidates in **JAM 2022 (Physics)**.
- Qualified for the **Integrated PhD interview** at **IISER Thiruvananthapuram** (2022).

- Ranked first in the B.Sc. Physical Science programme, University of Delhi (2021).
- **INSPIRE Scholar (2018–2023)**: ranked in the top 1% of 2.9 million students in Uttar Pradesh, India.

## Schools, Workshops and Conferences

- Accepted invitation to participate in the **Bangkok School on Cosmology** & workshop at **Dark Side of the Universe**, Mandarin Hotel Bangkok, Thailand (Organized by ICTP, Italy & Chulalongkorn University, Bangkok), 06 December –18 December 2026.
- Participated in **Cosmology from Home 2026**, 23 September –3 October 2026 in online mode. [link]
- Participated in the **School on the Concurrence of Mega-Surveys**, International Center for Theoretical Sciences (ICTS), Bengaluru, India, 27 July–4 August 2026 [link].
- Participated in the **Summer School on Cosmology**, International Center for Theoretical Physics (ICTP), Trieste, Italy, 29 June–10 July 2026 [Certificate].
- Participated in the **School on 21-cm Cosmology**, Tata Institute of Fundamental Research (TIFR), Mumbai, India, 3–6 March 2026 [Certificate].
- Participated in the **Workshop on 21-cm Cosmology**, Center for Strings, Gravitation, and Cosmology, Department of Physics, IIT Madras, India, 9–13 December 2024 [Certificate].
- Served as a **volunteer organizer** for the **Conference on Contemporary Physics**, IIT Kharagpur, held on 29–31 March 2026 as part of the Institute's Platinum Jubilee celebrations.

## Memberships

- Active member of the SKAO **Our Galaxy Science Working Group** (Dec 2024–Present). [Link]
- Active member of the SKAO **Epoch of Reionization Science Working Group** (Dec 2024–Present). [Link]
- Member of the **American Physical Society** (2022–Present).

## Technical Skills

- **Programming Languages:** Python, C++, C, Fortran, Julia, Bash, IDL.
- **Scientific Libraries and Frameworks:** NumPy, SciPy, Matplotlib, Astropy, Healpy, h5py, emcee, CosmoMC, CLASS, CAMB, Nbodykit.
- **Machine Learning and Probabilistic Modeling:** TensorFlow, PyTorch, JAX, scikit-learn
- **Astronomical and Cosmological Tools:** `TOPCAT`, `DS9`, `Aladin`, `IRAF`, `GILDAS`, Planck Legacy Archive tools.
- **Data Formats and Standards:** FITS, HDF5, CSV.
- **Platforms and Scientific Software:** Jupyter Notebook, MATLAB, LaTeX, Git, Linux/Unix, SLURM (HPC job scheduler).

## Certifications

**Remote Sensing & Digital Image Analysis — Global Navigation Satellite Systems**
*Indian Institute of Remote Sensing (IIRS), Dehradun, India*

- Developed expertise in **Synthetic Aperture Radar (SAR)** data analysis (NASA missions), including multi-frequency band processing.
- Investigated **active microwave remote sensing** techniques, focusing on backscatter analysis for Earth observation.
- Applied **thermal infrared (TIR) imagery** and **Advanced ATSR data** (used by ISRO) to study:
  - Land surface temperature (LST) dynamics
  - Monsoon pattern analysis
  - Pollution monitoring and environmental assessments

## Extracurricular Activities

- Participated in **Manthan** initiative (Govt. of India + Industry + Academia).
- Interests: Chess, swimming, meditation, reading books.